

Probability and Statistics

Unit VI: Linear Statistical Models

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Correlation

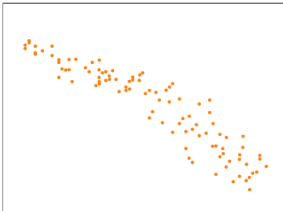
Correlation

- Correlation is a statistical measure (expressed as a number) that describes the size and direction of a relationship between two or more variables.
- Two variables are said to be correlated if change in one variable affects the change in other variable, and the relation between them is known as correlation.

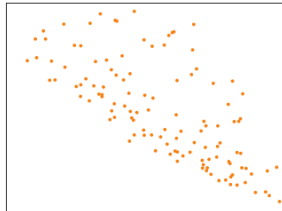
Correlation

FIVE CORRELATIONS

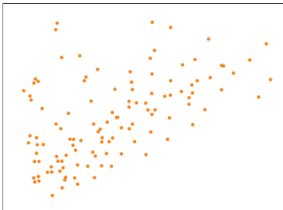
(a) Strong Negative Correlation ($r = -.933$)



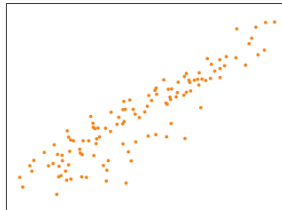
(b) Moderate Negative Correlation ($r = -.674$)



(c) Moderate Positive Correlation ($r = .518$)



(d) Strong Positive Correlation ($r = .909$)



Karl Pearson's Product Moment coefficient of correlation:

Correlation coefficient between two random variables X and Y, usually denoted by $r(X, Y)$ or r_{xy} and defined as

$$r(X, Y) = \frac{\frac{\sum x_i y_i}{n} - \bar{x} \cdot \bar{y}}{\sqrt{\frac{\sum x_i^2}{n} - \bar{x}^2} \cdot \sqrt{\frac{\sum y_i^2}{n} - \bar{y}^2}}$$

- If $r = +1$ then correlation is perfectly positive,
- If $r = -1$ then correlation is perfectly negative,
- If $r = 0$ then variables are uncorrelated.

$$r(X, Y) = \frac{n \sum XY - \sum X \sum Y}{\sqrt{n \sum X^2 - (\sum X)^2} \cdot \sqrt{n \sum Y^2 - (\sum Y)^2}}$$

Question

Determine the value of the coefficient of correlation r , for the following data:

X	4	6	7	11	14	17	21
Y	18	12	13	8	7	7	4

Karl Pearson Coefficient of correlation is given by:

$$r = \frac{n \sum XY - \sum X \sum Y}{\sqrt{n \sum X^2 - (\sum X)^2} \cdot \sqrt{n \sum Y^2 - (\sum Y)^2}}$$

$$\begin{aligned}n &= 7, \sum X = 80, \sum Y = 69, \sum XY = 624, \\ \sum X^2 &= 1148, \sum Y^2 = 815, \\ r &= -0.92698\end{aligned}$$

Question

Calculate the correlation coefficient for the following heights (in inches) of fathers (X) and their sons(Y)

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

$$r(X, Y) = \frac{\frac{\sum x_i y_i}{n} - \bar{x} \cdot \bar{y}}{\sqrt{\frac{\sum x_i^2}{n} - \bar{x}^2} \cdot \sqrt{\frac{\sum y_i^2}{n} - \bar{y}^2}}$$

Question

A computer while calculating correction coefficient between two variables X and Y from 25 pairs of observations obtained the following results:

$$n = 25, \sum X = 125, \sum X^2 = 650, \\ \sum Y = 100, \sum Y^2 = 460, \sum XY = 508$$

$$r(X, Y) = \frac{n \sum XY - \sum X \sum Y}{\sqrt{n \sum X^2 - (\sum X)^2} \cdot \sqrt{n \sum Y^2 - (\sum Y)^2}}$$

Spearman's Rank Correlation:

The method developed by Spearman is simpler than Karl Pearson's method since, it depends upon ranks of the items and actual values of the items are not required.

Hence this can be used to study correlation even when actual values are not known. For instance, we can study correlation between intelligence and honesty by this method.

$$R = 1 - \frac{6 \sum d_i^2}{n^3 - n}$$

$$d_i = R_1 - R_2$$

R_1 : rank of X

R_2 : rank of Y

Question

Calculate the rank correlation coefficient from the following data.

X	1	3	7	5	4	6	2	10	9	8
Y	3	1	4	5	6	9	7	8	10	2

X	1	3	7	5	4	6	2	10	9	8
Y	3	1	4	5	6	9	7	8	10	2
d_i										
d_i^2										

We know,

$$R = 1 - \frac{6 \sum d_i^2}{n^3 - n}$$

Question

Six students got the following Mathematics and Physics:

<i>Maths</i>	78	36	98	25	75	82
<i>Physics</i>	84	51	91	60	68	62

Calculate the rank correlation coefficient.

[illegible]

Ranks are repeated

If ranks are repeated then the Spearman's Rank correlation formula becomes

$$R = 1 - \frac{6[\sum_i^n d_i^2 + \sum_{i=1}^n \frac{m_i^3 - m_i}{12}]}{n^3 - n}$$

Where **m** is the number of times an item is repeated

Question

The following table shows the marks obtained by 10 students in Accountancy and Statistics. Find the Spearman's coefficient of rank correlation.

No.	1	2	3	4	5	6	7	8	9	10
Acc	45	70	65	30	90	40	50	57	85	60
Stat	35	90	70	40	95	40	60	80	80	50

$$R = 1 - \frac{6[\sum d_i^2 + \sum_{i=1}^n \frac{m_i^3 - m_i}{12}]}{n^3 - n}$$

Acc	R_1	Stats	R_2	d_i	d_i^2
45		35			
70		90			
65		70			
30		40			
90		95			
40		40			
50		60			
57		80			
85		80			
60		50			

- ① Line of regression of y on x is:

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

where regression coefficient of y on x is given by

$$b_{yx} = \frac{\text{Cov}(x,y)}{\sigma_x^2} = r \frac{\sigma_y}{\sigma_x}$$

- ② Line of regression of x on y is:

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

where regression coefficient of x on y is given by

$$b_{xy} = \frac{\text{Cov}(x,y)}{\sigma_y^2} = r \frac{\sigma_x}{\sigma_y}$$

Properties:

- 1 Lines of regression passes through the point $(\bar{x}, \bar{y})$
- 2 $b_{yx}b_{xy} = r^2$. $[r = \sqrt{b_{yx} \cdot b_{xy}}]$
- 3 b_{yx}, b_{xy} have same sign.

Question

*In partially destroyed laboratory record of an analysis of correlation data, the following results only are legible- Variance of $X = 9$, regression equations are: $8X - 10Y + 66 = 0$ & $40X - 18Y = 214$
What was*

- 1 the mean of X and Y
- 2 the correlation between X and Y
- 3 the S.D. of Y

Least Squares Straight Line

For a given set of N data points $(x_1, y_1), (x_2, y_2), \dots (x_N, y_N)$
assume that the straight line

$$Y = a_0 + a_1 X$$

fits to the data in the least squares sense
Normalized equation are given by

$$\sum Y_i = Na_0 + a_1 \sum X_i$$

$$\sum X_i Y_i = a_0 \sum X_i + a_1 \sum X_i^2$$

known as "**Normal equations**".

Question

Obtain the equation of the line of regression of cost on age from the following table giving the age of a car of certain make and the annual maintenance cost.

Age of car(in years)	2	4	6	8
Maintenance (in thousand of Rs.)	5	7	8.5	11

Also find maintenance cost of the car if its age is 9 years

[Ans : $y = 3 + 0.975x$ and $y = \text{Rs. } 11775$]

Question

It is given that the means of x and y are 5 and 10. If the line of regression of y on x is parallel to the line $20y = 9x + 40$, estimate the value of y at $x = 30$

[Ans: $20y = 9x + 155$ and $y = 21.25$]

Question

Find the two lines of regression from the following data

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

[Ans : $x = 30.364 + 0.545 y$ and $y = 23.667 + 0.667 x$]

